A380X weather radar GAIN, ELEVN and TILT

The manual GAIN, ELEVN and TILT settings of the MFD SURV CONTROLS page reach the ND messages, the VD white line and the radar picture itself (GAIN moves the colour levels, a TILT or ELEVN in the ground cuts the weather), and start from their FCOM default values

Aircraft	A380X
Where to find it	MFD SURV > CONTROLS (WXR block: GAIN, ELEVN/TILT); ND weather picture and message (right side above the TERR box); VD
Branch	`core/a380x-wxr-controls`, based on `core/nd-weather-terrain` with `core/a380x-mfd-fms` merged in (stacked: needs core/nd-weather-terrain, core/surveillance and core/a380x-mfd-fms merged first). The terrain view resend fix of comes with its base `core/nd-weather-terrain`.
Status	ND messages tested in the sim by the user: OK; FCOM defaults user-checked in the sim; VD white line tested in the sim by the user: OK. GAIN acting on the radar picture tested in the sim by the user: OK; TILT / ELEVN cutting the picture not sim-tested (left out of that test flight); native test wxr_controls_test.
Pictures	No pictures: the messages and the line are drawn by the stock ND / VD code (no screenshot of this state).

1. ND messages of the manual GAIN, ELEVN and TILT
2. GAIN, TILT and ELEVN take the FCOM defaults at the first selection
3. VD white line of the manual ELEVN and TILT
4. GAIN and ELEVN/TILT change the radar picture

1. ND messages of the manual GAIN, ELEVN and TILT

> **Aircraft:** A380X | **Where:** ND, right side above the TERR box | **Status:** Tested in the sim by the user: OK.

What it is for

The GAIN button and the ELEVN/TILT option list of the MFD SURV CONTROLS page kept their state but showed nothing. The ND now shows the manual radar settings as the A380 FCOM WXR MESSAGES (DSC-34-20-30-20 P 17).

What you see

SURV CONTROLS setting	ND message
GAIN AUTO, ELEVN/TILT AUTO	**WX** in green (as before)
GAIN MAN, value 53	**GAIN 53%** in cyan, above the mode line; with ELEVN/TILT at AUTO the mode line is empty and GAIN takes its place
ELEVN/TILT at TILT, value -12.0	**TILT -12.0°** in cyan instead of WX
ELEVN/TILT at ELEVN, value 12500, QNH reference	**ELEVN 12500FT** in cyan instead of WX
ELEVN/TILT at ELEVN, STD baro reference on that side	**ELEVN FL125** in cyan
Manual mode without a value	GAIN ---%, TILT ---.-°, ELEVN -----FT (or ELEVN FL---)
MODE MAP	MAP in green (GAIN nn% above it with GAIN MAN); no ELEVN / TILT message
WXR OFF	WXR OFF in white, no GAIN line

Limitations and why

Limitation	Why	What to do
GAIN, ELEVN and TILT act on the radar picture only in part.	The MSFS radar has no gain, no cell heights and no tilt; tilting its beam crashed the gauge.	GAIN moves the colour levels; ELEVN / TILT only hide the weather where they reach the ground (see the last feature).
Cyan is a choice.	The A380 FCOM gives no colour; cyan is the colour of the manual modes of the same radar family in the A320 FCOM (DSC-34-SURV-30-30 P 10).	-
The grey background the FCOM figure prints behind GAIN is not drawn.	Not reproduced.	-

How to use

1. MFD **SURV > CONTROLS**, WXR block (WXR AUTO, MODE WX).
2. GAIN: press the button to **MAN**, enter a value in the field below it (0 to 100 %).
3. ELEVN/TILT: select **ELEVN** (enter an altitude or FL) or **TILT** (enter -15.0 to +15.0 degrees) in the field below the list.
4. Read the message on the ND (WX overlay selected). Back to **AUTO** for WX.

How to test

#	Do this	Expected result
1	A380X, CPT EFIS **WX** overlay, ND ARC. MFD SURV > CONTROLS: WXR AUTO, MODE WX, GAIN AUTO, ELEVN/TILT AUTO.	ND: **WX** in green on the right.
2	GAIN to **MAN**, enter **53**.	ND: **GAIN 53%** in cyan, the WX message gone.
3	ELEVN/TILT: select **TILT**, enter **-12**.	ND: **TILT -12.0°** in cyan, with GAIN 53% above it.
4	Select **ELEVN**, enter **12500** (QNH set on the CPT side).	ND: **ELEVN 12500FT** in cyan.
5	Set the CPT baro reference to **STD**.	ND: **ELEVN FL125**.
6	GAIN and ELEVN/TILT back to **AUTO**.	ND: **WX** in green, no GAIN line.
7	**MODE MAP** with GAIN MAN.	ND: **MAP** in green with the GAIN line above it.
8	WXR **OFF** (confirm).	ND: **WXR OFF** in white, no GAIN line.

2. GAIN, TILT and ELEVN take the FCOM defaults at the first selection

> **Aircraft:** A380X | **Where:** MFD SURV > CONTROLS, WXR block | **Status:** User-checked in the sim after a flight reload: GAIN 50 %, TILT +3.0° on the ground, ELEVN 100 FT at CYUL.

What it is for

A380 FCOM DSC-34-20-30-20-10 (GAIN and ELEVN knobs): "The default GAIN value is 50 %", "The default ELEVN value is the current aircraft altitude", the default TILT value is +3.00° on ground and 0° in flight. The page applies a default only when nothing is entered, but a value nobody had written read 0, a valid entry: after a power-up the first GAIN MAN, TILT or ELEVN selection kept 0. The MFD now starts with the three settings at no entry.

What you should see

Situation	What you should see
First GAIN MAN after the MFD starts	GAIN field 50 %; ND GAIN 50%.
First TILT selection on the ground	TILT +3.0°; ND TILT +3.0°.
First TILT selection in flight	TILT +0.0°.
First ELEVN selection	The current altitude, rounded to 100 ft (FL with the STD reference).
Later selections	The value entered or set before is kept.

How to use

1. Select GAIN **MAN**, **ELEVN** or **TILT** on the SURV CONTROLS page: the default value is entered.
2. Overwrite it in the entry field if needed.

How to test

#	Do this	Expected result
1	A380X on the ground, after a flight load or reload. MFD SURV > CONTROLS: GAIN to **MAN**.	GAIN field **50 %**; ND (WX overlay) **GAIN 50%**.
2	ELEVN/TILT: select **TILT**.	TILT field **+3.0°**; ND **TILT +3.0°**.
3	Select **ELEVN**.	ELEVN field = the airport elevation rounded to 100 ft (for example 100 FT at CYUL).
4	Enter TILT -5, go to AUTO, then select TILT again.	TILT **-5.0°** is kept (no default over an entry).
5	Reload the flight, take off, and select TILT for the first time in flight.	TILT **+0.0°**.

3. VD white line of the manual ELEVN and TILT

> **Aircraft:** A380X | **Where:** VD under the ND | **Status:** Tested in the sim by the user: OK.

What it is for

A380 FCOM DSC-34-20-30-20: "The VD displays a white line indicating the elevation or tilt value". In manual ELEVN the VD shows a horizontal white line at the selected altitude; in manual TILT a straight white line from the aircraft symbol at the selected angle (zero tilt = the horizon as seen by the ADIRS), as in the FCOM VD figures.

What you should see

Situation	What you should see
ELEVN selected	Horizontal white line at the selected altitude, from the aircraft to the end of the VD range.
TILT selected	Straight white line from the nose of the aircraft symbol, up for a positive tilt, down for a negative one.
Selected altitude off the VD scale	No line.
ELEVN/TILT AUTO, MODE MAP, WXR OFF or no WX overlay	No line (it goes with the ELEVN / TILT message of the ND).

Limitations and why

Limitation	Why	What to do
The line shows only while that side's ND shows the WX function.	Design choice: the line goes with the ND message.	-
The tilt line is straight.	Design choice, as in the FCOM figure: no earth curvature or beam refraction.	-
The weather on the VD does not follow ELEVN or TILT.	The MSFS radar has no height data; the ground cut of a manual ELEVN / TILT applies to the ND only. The VD weather colours follow the GAIN.	-

How to use

1. Select the **WX** overlay on the EFIS panel and ND ARC or ROSE NAV (the pages with the VD).
2. On MFD SURV > CONTROLS select **ELEVN** or **TILT** and enter the value.
3. Read the white line on the VD.

How to test

#	Do this	Expected result
1	A380X in flight, for example at 10 000 ft. CPT EFIS WX overlay, ND ARC 40 NM. SURV CONTROLS: ELEVN/TILT **ELEVN**, enter **12000**.	VD: a horizontal white line at 12 000 ft, ahead of the aircraft symbol.
2	Select **TILT**, enter **+3**.	VD: a straight white line from the nose of the aircraft symbol, rising to the end of the range.
3	Enter **-3**.	The line goes down from the aircraft symbol.
4	Select **ELEVN** with a value far above the VD scale (for example 60000).	No line.
5	ELEVN/TILT back to **AUTO**.	The line goes; ND shows WX.
6	With TILT selected, deselect the WX overlay.	The line goes with the ND message.

4. GAIN and ELEVN/TILT change the radar picture

> **Aircraft:** A380X | **Where:** ND weather picture; MFD SURV > CONTROLS, WXR block | **Status:** GAIN tested in the sim by the user: OK (FL150, ARC 40 NM: AUTO calibrated, 0 % sparse green, 100 % red cores, AUTO back to the calibrated picture, the VD following). TILT and ELEVN cutting the picture: not sim-tested. Native test wxr_controls_test (ndwxr).

What it is for

The manual GAIN, TILT and ELEVN entries changed the ND messages and the VD line only. They now act on the radar picture, within what the MSFS radar allows: its views take a table of rain rate bands, but have no cell heights and no tilt.

- **GAIN** moves the rain rate each colour starts at, as the receiver gain moves the reflectivity of each colour (A380 FCOM DSC-34-20-30-10/-20; Honeywell RDR-4000 Pilot manual: MIN about -16 dBZ, MAX about +10 dBZ, calibrated in AUTO). 50 %, the FCOM default, is the calibrated gain (the same picture as AUTO); 0 % is the MIN, 100 % the MAX. A lower gain turns red to yellow, yellow to green and removes green (Airbus Safety First #22); a higher gain the opposite. Turbulence (magenta) does not change.

- A manual **TILT** down shows the weather only up to the range where the tilted beam reaches the ground (the radar removes the ground returns with the TAWS terrain, DSC-34-20-30-10). A level or up tilt changes nothing.
- A manual **ELEVN** at or below the ground under the aircraft shows no weather; above it, the full picture.

What you should see

Situation	What you should see
GAIN AUTO, or MAN at 50 %	The calibrated picture (as before).
GAIN MAN 0 %	Red cells shrink or turn yellow, yellow turns green, light green disappears.
GAIN MAN 100 %	More and larger green, yellow and red areas.
TILT -10.0 at about 10 000 ft above the ground	Weather only inside about 9 NM.
TILT -3.0 at about 10 000 ft above the ground	Weather out to about 34 NM.
ELEVN at or below the ground elevation	No weather on the ND.
VD	Its weather colours follow the GAIN; the TILT / ELEVN cut is on the ND only.

Limitations and why

Limitation	Why	What to do
ELEVN and TILT cannot show a slice above or below the cells.	MSFS gives no cell tops: only the part in the ground can be removed.	-
The ground ahead is taken level at the elevation under the aircraft.	The gauge cannot read the terrain ahead back (design choice). The beam uses the earth curvature with the 4/3 radius of the standard refraction.	-
The GAIN scale is a design choice.	The FCOM gives the percentage but not its scale: 50 % calibrated, 0 % and 100 % the RDR-4000 MIN and MAX, linear in dB on each side.	-
A GAIN change may repaint the picture over a few seconds.	Not measured yet: the engine may repaint it sector by sector with its sweep.	-

How to use

1. MFD **SURV > CONTROLS**, WXR block (WXR AUTO, MODE WX).
2. GAIN **MAN** and a value: below 50 % to look into heavy cells (less colour), above 50 % to see weak returns. Back to **AUTO** for the calibrated picture.
3. ELEVN/TILT: **TILT** with a down angle (or **ELEVN**) to keep the weather only where the beam is above the ground. Back to **AUTO** for the full picture.

How to test

#	Do this	Expected result
1	A380X in flight at about FL100 (about 10 000 ft above the ground) with rain cells ahead. CPT EFIS: WX on, ND ARC 40 NM. MFD SURV > CONTROLS: WXR AUTO, MODE WX, TURB AUTO. Screenshot of the ND.	The calibrated picture.
2	GAIN button to **MAN** (field 50 %); wait 10 s.	Same picture as before.

3	Enter **0** in the GAIN field; wait 10 s.	Red cells shrink or turn yellow, yellow turns green, light green disappears. Magenta unchanged.
4	Enter **100**; wait 10 s.	More and larger green, yellow and red areas than at the start.
5	GAIN back to **AUTO**.	The first picture again. The VD weather colours followed each step.
6	ELEVN/TILT: select **TILT** (field +0.0).	The picture does not change.
7	Enter **-10**; wait 5 s.	Weather only inside about 9 NM; ND message TILT -10.0.
8	Enter **-3**; wait 5 s.	Weather out to about 34 NM; ND message TILT -3.0.
9	Over land at least 500 ft high: select **ELEVN**, enter 0 (FL 000 with STD); then 12000 (FL 120).	At 0: no weather on the ND. At 12000: the full picture.
10	ELEVN/TILT back to **AUTO**.	The full picture again.